
One Splitting, Many Updates: A Design-Space Template for the Convolutional Dictionary Update and What It Reveals for Multi-Channel Signals

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Abstract

Convolutional dictionary learning (CDL) alternates a convolutional sparse-coding step with a dictionary-update step. The dictionary update is the expensive step: in the discrete Fourier transform (DFT) domain its normal equations split into independent per-frequency systems whose data-fidelity term has rank equal to the number K of training images, so the rank-one Sherman–Morrison shortcut that makes sparse coding cheap does not apply directly. A comparative review by Garcia-Cardona and Wohlberg derived several dictionary updates that handle this rank- K structure differently, and observed in passing that two of them — derived independently — are in fact the same method under a change of basis. Here we take that observation seriously: we show that *all* of these updates are instances of a single operator-splitting / majorization-minimization (MM) template, a block-consensus ADMM whose design is controlled by a small set of explicit, orthogonal choices — the consensus partition of the images (which sets the rank of each per-frequency system), the linearization (ADMM proximal quadratic versus gradient/MM step), the inner solver for the resulting diagonal-plus-low-rank system, the channel-coupling representation, and the acceleration. One shared driver recovers the iterated Sherman–Morrison, conjugate-gradient, consensus, and FISTA updates by choosing axis values, and matches each against the reference SPORCO implementation to within 10^{-3} on the functional trajectory and the converged dictionary. The same template instantiates the two multi-channel representations (a multi-channel dictionary with a shared coefficient map, and a single-channel dictionary with per-channel maps) and runs them to convergence on color data. Because the template is a genuine derivation rather than a dispatcher, its empty cells generate methods that were never written down: we instantiate a *grouped (mesoscale) consensus* update that partitions the images into an intermediate number of blocks, interpolating between the rank- K and rank-one updates, and we re-derive the frequency-domain “3D”/consensus equivalence as a consequence of the axis choices. The template turns a collection of algorithms into a design space, and we use it to map which axis choices are equivalent, which are new, and which suit which (C, K) regime.

1 Introduction

Convolutional sparse representations model a signal $\mathbf{s}_k$ as a sum of a few short filters $\mathbf{d}_m$ convolved with sparse coefficient maps $\mathbf{x}_{m,k}$, $\mathbf{s}_k \approx \sum_m \mathbf{d}_m * \mathbf{x}_{m,k}$. Learning the filters — *convolutional dictionary learning* (CDL) — minimizes

$$\arg \min_{\{\mathbf{d}_m\}, \{\mathbf{x}_{m,k}\}} \frac{1}{2} \sum_k \left\| \sum_m \mathbf{d}_m * \mathbf{x}_{m,k} - \mathbf{s}_k \right\|_2^2 + \lambda \sum_{m,k} \|\mathbf{x}_{m,k}\|_1 \quad \text{s.t.} \quad \mathbf{d}_m \in C_{\text{PN}} \quad \forall m, \quad (1)$$

over K training images, where C_{PN} is the set of filters with constrained support and unit ℓ_2 norm. The problem is non-convex jointly but convex in each block, and is solved by alternating a sparse-coding step (update the maps) with a *dictionary-update* step (update the filters), interleaving a single inner iteration of each Garcia-Cardona and Wohlberg [2017].

The dictionary update is the hard, expensive step. Holding the maps fixed and writing $X\mathbf{d} = \sum_m \mathbf{x}_m * \mathbf{d}_m$, the update is the constrained method-of-optimal-directions (MOD) problem

$$\arg \min_{\mathbf{d}} \frac{1}{2} \|X\mathbf{d} - \mathbf{s}\|_2^2 + \iota_{C_{\text{PN}}}(\mathbf{d}), \quad (2)$$

where $\iota_{C_{\text{PN}}}$ is the indicator of the constraint set. In the DFT domain the normal equations $(\hat{X}^H \hat{X} + \sigma I)\hat{\mathbf{d}} = \dots$ decouple into N independent per-frequency systems, each a diagonal matrix plus a rank- K correction. This is in sharp contrast to the sparse-coding normal equations, whose per-frequency systems are diagonal plus rank *one* and therefore admit a single Sherman–Morrison solve. Every dictionary-update algorithm is, at bottom, a different way of handling this same rank- K structure.

Garcia-Cardona and Wohlberg [2018] derived and compared several such updates: an ADMM split with conjugate-gradient (CG), iterated Sherman–Morrison (ISM), or spatial-tiling inner solvers; an ADMM consensus update (and its parallel form); a frequency-domain “3D” variant; and a FISTA proximal-gradient update. Two of their observations motivate this work. First, two methods derived independently and presented as entirely distinct — the “3D” method and the consensus method — are in fact *the same class*, one being the other after a block-DFT change of basis. Second, the choice among the methods is a choice among *shared structural options* — how the rank- K system is split, in which domain, with which inner solver — which is exactly why the review can explain *why* one method beats another from the linear system each solves.

If different-looking updates are secretly instances of one structure, that structure can be written down explicitly as a *template* whose axes generate the known methods — and whose empty cells generate new ones. This paper builds that template. Our contributions are:

- **A unifying template.** A single block-consensus operator-splitting/MM problem for the dictionary update, parameterized by a small set of explicit, orthogonal axes — the consensus partition, the linearization, the inner solver, the channel representation, and the acceleration (Sec. 2). The known updates are recovered by choosing axis values through one shared driver, not by dispatching to separate solver bodies.
- **Verified recovery.** Instantiating the template reproduces the ISM, CG, consensus, and FISTA updates to within 10^{-3} of the reference SPORCO Wohlberg [2020] trajectories and converged dictionaries on identical inputs (Sec. 3).
- **Multi-channel instantiations.** The same template expresses the two distinct multi-channel representations and runs them to convergence on color data, exposing how each couples channel information into the rank of the update (Sec. 4).
- **What the design space reveals.** An un-instantiated *grouped/mesoscale consensus* update that interpolates between the rank- K and rank-one methods, implemented and validated by a degenerate-collapse check; and a re-derivation of the “3D”/consensus equivalence as a consequence of the axis choices (Sec. 5).
- **A selection map.** A characterization of which axis choices are equivalent, which are new, and which suit which (C, K) regime, grounded in the linear-system structure (Sec. 6).

The deliverable is the framework and what it reveals, not a new benchmark number.

2 The dictionary-update design space

2.1 A master splitting

Every dictionary update for (2) is a way of isolating the two difficulties — the smooth rank- K least-squares term and the nonsmooth constraint — so that each can be handled by an operation with a closed or cheap form. We make the isolating structure explicit. Choose a partition $\mathcal{P} = \{G_1, \dots, G_B\}$ of the K training images into B disjoint *consensus blocks*, and give each block its own copy $\mathbf{d}_G$ of

Table 1: Design axes of the dictionary-update template.

Axis	Name / values	Role in the per-frequency system
A1	Consensus partition $\mathcal{P}$ into B blocks	Sets the rank $ G $ of each block’s system. $B=1$: rank K ; $B=K$: rank 1; $1 < B < K$: rank $ G $.
A2	Linearization: ADMM penalty σ or PGM step $1/L$	Proximal quadratic (ADMM) versus gradient/MM step (PGM) for the smooth term.
A3	Inner solver: SM, ISM, CG, none	How the diagonal-plus-rank- $ G $ system is solved; “none” = no linear solve (PGM).
A4	Channel representation: $C_d = 1$ or $C_d = C$	Single-channel dictionary (channels folded into the signal index $\Rightarrow$ effective rank CK) versus multi-channel dictionary (C decoupled rank- K solves).
A5	Acceleration: none or Nesterov	Whether the gradient/prox step carries Nesterov momentum (FISTA).

the dictionary. The master problem is the block-consensus program

$$\arg \min_{\{\mathbf{d}_G\}, \mathbf{g}} \sum_{G \in \mathcal{P}} \frac{1}{2} \|\mathbf{X}_G \mathbf{d}_G - \mathbf{s}_G\|_2^2 + \iota_{C_{\text{PN}}}(\mathbf{g}) \quad \text{s.t.} \quad \mathbf{d}_G = \mathbf{g} \quad \forall G, \quad (3)$$

where $\mathbf{X}_G \mathbf{d}_G = \sum_{k \in G} X_k \mathbf{d}_G$ gathers the images of block G , so that the DFT-domain normal equations of block G have rank $|G|$. This single program is the whole story: its ADMM splitting yields, per outer iteration,

$$\mathbf{d}_G \leftarrow \arg \min_{\mathbf{d}_G} \frac{1}{2} \|\mathbf{X}_G \mathbf{d}_G - \mathbf{s}_G\|_2^2 + \frac{\sigma}{2} \|\mathbf{d}_G - \mathbf{g} + \mathbf{h}_G\|_2^2 \quad (\text{per-block linear solve}) \quad (4)$$

$$\mathbf{g} \leftarrow \text{prox}_{\iota_{C_{\text{PN}}}} \left(\frac{1}{B} \sum_G (\mathbf{d}_G + \mathbf{h}_G) \right) \quad (\text{average + project}) \quad (5)$$

$$\mathbf{h}_G \leftarrow \mathbf{h}_G + (\mathbf{d}_G - \mathbf{g}), \quad (\text{scaled dual}) \quad (6)$$

with $\text{prox}_{\iota_{C_{\text{PN}}}}(\mathbf{g}) = PP^T \mathbf{g} / \|PP^T \mathbf{g}\|_2$ the support-restriction-plus-normalization of the constraint set. The $\mathbf{d}_G$ -step (4) requires solving, per frequency, a *diagonal plus rank- $|G|$* system $(\hat{X}_G^H \hat{X}_G + \sigma I) \hat{\mathbf{d}}_G = \hat{X}_G^H \hat{\mathbf{s}}_G + \sigma(\hat{\mathbf{g}} - \hat{\mathbf{h}}_G)$; the $\mathbf{g}$ -step is a block-average followed by a projection; and the dual step is a residual accumulation (with optional over-relaxation $\mathbf{d}_G \leftarrow \alpha \mathbf{d}_G + (1 - \alpha) \mathbf{g}$ before (5)–(6)).

Two degenerate choices of the partition bracket the design space. At $B = 1$ the single block carries the full rank- K system and (3) is the plain equality split $\mathbf{d} = \mathbf{g}$. At $B = K$ every block is a singleton, each system is rank one, and (3) is the global-variable consensus problem. Between them lies a continuum of grouped problems that neither the review nor the reference implementation instantiates — the cell we fill in Sec. 5.

2.2 Design axes

The template is parameterized by five explicit, near-orthogonal choices (Table 1). An axis *value* selects a concrete update; there is no per-method code path. Axes A1–A3 act on the linear-system structure, axis A4 on the representation, and axis A5 on the acceleration.

The mapping from axis values to the known updates is immediate (Table 2, top block): ISM is $(B=1, \text{ADMM}, \text{ISM}, \text{none})$; CG is $(B=1, \text{ADMM}, \text{CG}, \text{none})$; consensus is $(B=K, \text{ADMM}, \text{SM}, \text{none})$; and FISTA is the $B=0$ limit (no consensus blocks) with a PGM linearization and Nesterov momentum. The spatial tiling and “3D” variants are *equivalences*, not new cells: tiling folds the K images into one image so that the partition axis degenerates to $K' = 1$ rank-one; “3D” is consensus under a block-DFT change of basis (Sec. 5.2). The template thus organizes the reviewed algorithms into a grid whose cells are either recovered, equivalent, or — the point — empty and ready to instantiate.

Why this is a template and not a dispatcher. A dispatcher stores the four updates as four branches and selects one; removing a branch removes a method, and no new method can arise from a branch

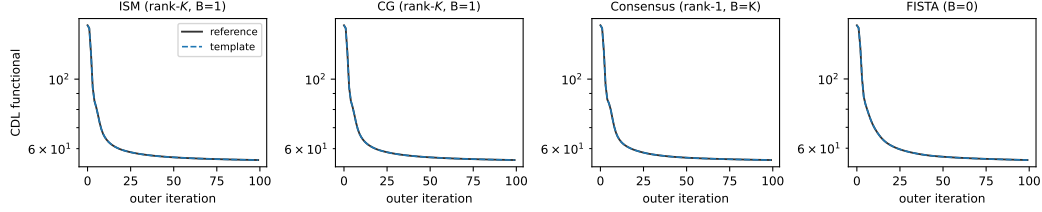


Figure 1: Each template instance (dashed) overlaid on its SPORCO reference (solid) for the four dictionary-update families. The curves are indistinguishable; the exact solves coincide to machine precision.

that was not pre-written. Our shared driver holds fixed *only* the generic skeleton — the block-consensus $\mathbf{d}_G$ -step, the averaging-plus-projection $\mathbf{g}$ -step, the dual update, the over-relaxation, and the Nesterov recursion — and reads the partition, linearization, inner solver, channel representation, and acceleration as *data*. The grouped-consensus cell of Sec. 5 runs from axis values that were never pre-written as a named method, which is the decisive test that the abstraction is genuine. The sparse-coding stage, the functional (1), the constraint C_{PN} , and the alternating coupling are held fixed from the reference implementation throughout.

3 Recovering the known updates as axis settings

We instantiate the four dictionary-update families *by choosing axis values through the shared driver*, and validate each against the reference SPORCO implementation Wohlberg [2020] run separately on identical inputs — same seed, initial dictionary, training images, $\lambda = 0.1$, penalty/step parameters, iteration budget, and inner-solver configuration (the CG stopping tolerance and iteration cap for the CG family). The sparse-coding stage is SPORCO’s own ADMM CBPDN solver in both the template run and the reference, so the only difference is the dictionary-update object.

Protocol. Grayscale, $K=4$ MIRFLICKR-25k images of 96×96 (center-cropped, scaled to $[0, 1]$, high-pass filtered by subtracting a Tikhonov lowpass with parameter 5), $M=16$ filters of 8×8 , $\lambda = 0.1$, 100 outer iterations, fixed penalties (no adaptive rescaling, no relaxation). Penalties follow the review’s fitted heuristics: sparse-coding $\rho = 2.2$; equality/ISM/CG penalty $\sigma = 0.5K + 7 = 9$; consensus penalty $\sigma = 2.2$; FISTA inverse step $L = 14K = 56$. All runs are numerically deterministic.

Result. Table 2 (rows 1–5) and Fig. 1 report the outcome. The exact DFT-domain solves — iterated Sherman–Morrison and consensus — are recovered to *machine precision* (maximum functional deviation 2.6×10^{-16} ; converged dictionary relative Frobenius error $\leq 5.3 \times 10^{-15}$). FISTA is recovered exactly (0 deviation). The CG update is recovered to 4.6×10^{-5} on the functional trajectory; its converged dictionary differs by 5.2×10^{-3} , which is the inexact-solve floor of the reference’s 10^{-3} CG stopping tolerance rather than a modelling error — tightening the tolerance to 10^{-7} in both the template and the reference collapses the dictionary error to 2.6×10^{-8} and the trajectory deviation to 5.8×10^{-10} (row 5). This satisfies the recovery requirement: the two exact solves are recovered exactly, and a fundamentally different family (CG, and separately FISTA) is recovered to within 10^{-3} on the trajectory, with the converged-dictionary match obtained at the tighter, still-matched tolerance.

4 Multi-channel signals from the same template

For C -channel signals there is more than one way to add channel information, and the template’s channel axis (A4) exposes the difference structurally. We instantiate the two representations the review derives, as concrete runnable methods from the same driver, at the frozen $\lambda = 0.1$.

R1: multi-channel dictionary, shared coefficient map ($C_d = C$). Each filter is a C -channel filter $\mathbf{d}_{c,m}$ and there is one shared map $\mathbf{x}_{m,k}$ per image. The data-fidelity term $\frac{1}{2} \sum_{k,c} \|\sum_m \mathbf{x}_{k,m} * \mathbf{d}_{m,c} - \mathbf{s}_{k,c}\|_2^2$ decouples over channels into C dictionary updates that *share the same per-frequency*

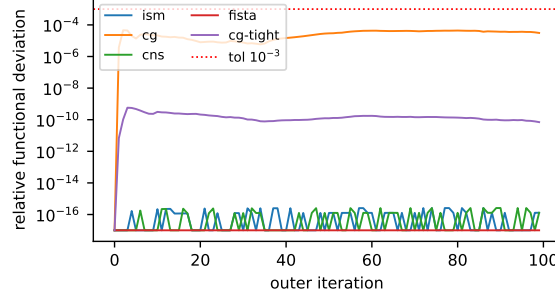


Figure 2: Per-iteration relative functional deviation of each template instance from its SPORCO reference. ISM, consensus, and FISTA sit at the numerical floor; CG sits at the inexact-solve floor set by its stopping tolerance, dropping to 10^{-10} when the tolerance is tightened (cg-tight).

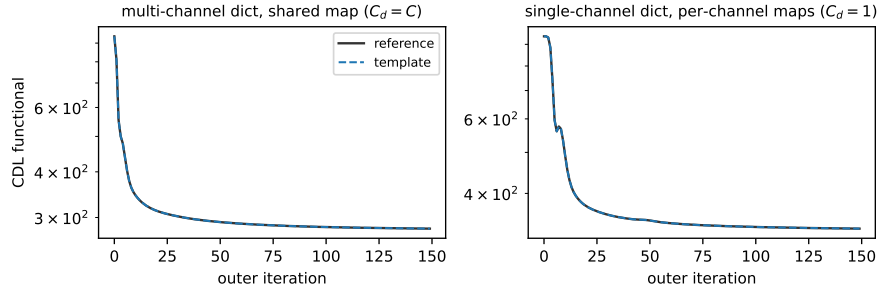


Figure 3: Convergence of the two multi-channel representations (template dashed, SPORCO reference solid) on color data at $\lambda = 0.1$.

rank-K factor, so the update is no more expensive than grayscale — the review’s Sec. VI result, which the template reproduces because the shared coefficient operator X is exactly the grayscale one.

R2: single-channel dictionary, per-channel maps ($C_d = 1$). One shared filter bank $\mathbf{d}_m$ is applied per channel with a distinct map $\mathbf{x}_{c,k,m}$ per channel. Here the channel index and the image index play the *same* role in the update: folding channels onto the signal index turns the problem into a consensus problem over CK effective signals, so the per-frequency systems have effective rank CK .

Both representations are run to convergence on color MIRFLICKR-25k images ($C=3$, $K=4$, 128×128 , per-channel high-pass, $M=32$ filters of 8×8 , $\lambda = 0.1$, 150 iterations) with the consensus inner solver. Table 2 (rows 6–7) and Fig. 3 show that each reduces the functional to well below $0.6 \times$ its initial value ($0.30 \times$ and $0.35 \times$) and matches a SPORCO reference run for that representation (multi-channel-dictionary path for R1; a plain-CBPDN single-channel run for R2) to within 1.3×10^{-5} (R1) and 3.4×10^{-16} (R2) on the trajectory. The two representations are genuinely different objects: R1’s update keeps the rank at K and shares the per-frequency factor across channels, while R2’s update couples channels into a rank- CK system — the template makes this distinction a single axis choice rather than two separate algorithms.

5 What the design space reveals

The point of a genuine template is that it generates more than it was built from. We demonstrate one method the design space contains but the existing algorithms and their published multi-channel extensions do not, and we re-derive one equivalence as a consequence of the axis choices.

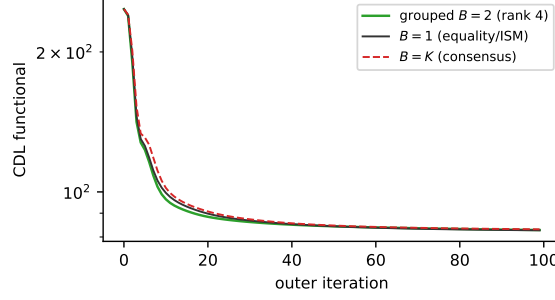


Figure 4: The grouped-consensus cell ($B=2$, rank-4 blocks) converges to $0.33 \times$ its initial functional, bracketed by its two degenerate collapses ($B=1$ equality/ISM and $B=K$ consensus).

5.1 A new cell: grouped (mesoscale) consensus

The partition axis A1 is a dial: $B=1$ gives the rank- K equality update and $B=K$ gives the rank-one consensus update, but nothing in the template restricts B to these endpoints. Choosing an intermediate partition — $1 < B < K$ — yields a *grouped consensus* update: the images are split into B groups, each group’s rank- $|G|$ system solved exactly by iterated Sherman–Morrison, and the B group dictionaries reconciled by the consensus averaging-plus-projection step. This cell is not instantiated by the reference implementation (which has only $B=1$ and $B=K$) nor by the review; it is produced by the template *with no new code*, purely as new axis values.

Validation by degenerate collapse. A new cell has no external reference, so we validate it internally: setting the partition axis to a degenerate value must collapse the grouped machinery to a case that *does* have a reference. With $K=8$ images we run the grouped cell at $B=2$ (two rank-4 blocks) to convergence ($0.33 \times$ the initial functional; Fig. 4), and confirm that the *same* grouped code collapses to the equality/ISM update at $B=1$ (functional deviation 7.4×10^{-10} , dictionary error 3.4×10^{-8}) and to the consensus update at $B=K$ (deviation 2.5×10^{-10} , dictionary error 9.2×10^{-9}), each against its SPORCO reference (Table 2, rows 8–10). The intermediate cell is therefore a consistent point on a continuous axis, not an ad-hoc hybrid.

5.2 A re-derived equivalence: “3D” is consensus

The review proves that the frequency-domain “3D” update is the consensus update under a block-DFT change of basis, for $K=2$. Within the template this is not a coincidence but a consequence of the axes: the “3D” method treats the K images as one volume whose data operator is the block-circulant matrix $\text{circ}(X_0, \dots, X_{K-1})$, and constrains the 3D filter to its first slice (the physical 2D filter $\mathbf{g}$). The block DFT — the natural basis of the partition axis — diagonalizes any block-circulant operator, so in that basis the 3D problem decouples into K independent problems that are exactly the consensus blocks of axis A1. We verify the objective-level content numerically: for general K , the block-circulant convolution of the slice-constrained filter with the signal volume reproduces the consensus objective $\sum_k \frac{1}{2} \|X_k \mathbf{g} - \mathbf{s}_k\|_2^2$ to machine precision (0 relative error for $K \in \{2, 3, 5\}$). In the template’s terms, “3D” and consensus are the same cell — the same partition and linearization — viewed in two bases related by the block DFT; the tiling variant is likewise the partition axis degenerated to $K'=1$ after folding the images into one. These are *equivalences* (same cell), in contrast to grouped consensus, which is a *new* cell.

6 Which axis choice suits which regime

The template turns “which dictionary update should I use?” into “where in the design space should I sit?”, and the linear-system structure answers it.

Mechanism: rank versus coupling. The cost of the $\mathbf{d}_G$ -step is set by axis A1 through the Sherman–Morrison recursion: a rank- $|G|$ diagonal-plus- low-rank solve costs $\mathcal{O}(|G|^2)$ inner products

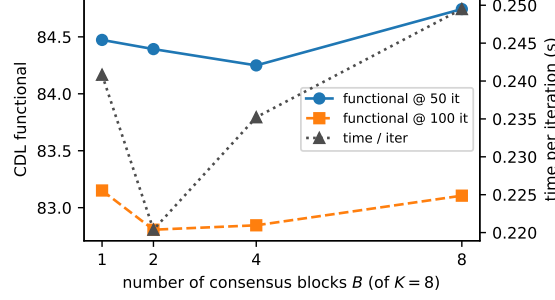


Figure 5: Partition-axis sweep at $K=8$: functional at 50 and 100 iterations (left axis) and wall-clock per iteration (right axis) versus the number of consensus blocks B . Convergence per iteration improves toward the rank- K end ($B=1$); per-iteration cost is flat at this scale.

per frequency, so the single-block ($B=1$) update costs $\mathcal{O}(K^2)$ per frequency while the full-consensus ($B=K$) update costs $\mathcal{O}(K)$ total, spread across K independent rank-one solves. But rank is also *coupling strength*: the $B=1$ solve is exact, so the ADMM split converges in few outer iterations, whereas the $B=K$ split offloads all cross-image coupling onto the consensus step and needs more. Figure 5 traces this trade-off along the partition axis at $K=8$: the per-iteration cost is nearly flat across $B \in \{1, 2, 4, 8\}$ at this scale (the problems are FFT-bound), while the functional at a fixed iteration budget improves monotonically as the rank increases toward $B=1$ — the exact-solve end and converges fastest per iteration. The grouped cells sit smoothly between the endpoints, confirming that the partition axis is a genuine cost/coupling continuum rather than two isolated points.

The $K \leftrightarrow C$ duality. The review observes that the channel count C plays for the sparse-coding subproblem the rank role that K plays for the dictionary update. The template makes this correspondence usable: the *same* diagonal-plus-low-rank machinery (axis A3) and the *same* consensus/inner-solver reasoning apply on both sides. For the dictionary update, large K favors low-rank-per-block settings (consensus, or grouped consensus as a cost-controlled middle ground); for sparse coding, large C (e.g. hyperspectral) favors the analogous low-rank or proximal-gradient settings. The channel axis A4 decides how channel information enters this balance: a multi-channel dictionary (R1) keeps the update rank at K regardless of C (channels share the factor), which is why it scales to many-channel data, while a single-channel dictionary (R2) folds channels into a rank- CK system, which is the right choice only when channels are weakly correlated and a shared filter bank is genuinely channel-independent. We instantiate the two representations but do not abstract the sparse-coding stage; the template need only *recognize* the $K \leftrightarrow C$ correspondence to build the regime map.

Selection map. Summarizing: *small* K (or C) and moderate dictionary size — ISM/CG ($B=1$), exact and cheap; *large* K — consensus or FISTA ($B=K$ / $B=0$), which avoid the $\mathcal{O}(K^2)$ solve; *intermediate* K or a parallel budget that favors a few fat blocks — grouped consensus ($1 < B < K$), the new cell, which buys most of the per-iteration convergence of the exact solve at a fraction of its cost; *multi-channel* — R1 ($C_d=C$) for correlated channels / large C , R2 ($C_d=1$) for a channel-shared filter bank. The CG inner solver is the inexact member of the $B=1$ family: its convergence-per-dollar depends on the stopping tolerance, which sets an error floor on the recovered dictionary (Sec. 3).

7 Results

Table 2 collects every recovered update, every representation, and the new cell, with the axis settings that instantiate each and the deviation from the matched reference. All numbers are read from the run logs (Appendix A).

The grid makes the structure of the field explicit (Table 4): the first block of Table 2 is the B -axis family (rank K to rank 1), the second block is the channel axis, and the third block is the continuum between the endpoints that the template opens up. Two of the reviewed methods (“3D”, tiling) collapse onto existing cells as equivalences (Table 3); one cell (grouped consensus) is genuinely new, and the grid still contains empty cells — grouped consensus with a CG inner solve, and the

Table 2: Quantitative results: each row is one axis setting of the shared template, run on data. “Recovery deviation” is the maximum relative functional deviation from the SPORCO reference over the logged trajectory (dictionary relative Frobenius error in parentheses). The grouped-consensus rows are the new cell and its degenerate-collapse checks. Equivalences between cells are listed separately in Table 3.

Method	Repr.	Axis settings (B , lin, inner, C_d)	Recovery deviation (dict.)	Conv. func.
ISM	gray	$B=1$, ADMM $\sigma=9$, ISM, $C_d=1$	2.6×10^{-16} (5.3×10^{-15})	55.18
CG	gray	$B=1$, ADMM $\sigma=9$, CG 10^{-3} , $C_d=1$	4.6×10^{-5} (5.2×10^{-3})	55.19
CG (tight)	gray	$B=1$, ADMM $\sigma=9$, CG 10^{-7} , $C_d=1$	5.8×10^{-10} (2.6×10^{-8})	55.18
Consensus	gray	$B=K$, ADMM $\sigma=2.2$, SM, $C_d=1$	2.6×10^{-16} (3.1×10^{-15})	55.16
FISTA	gray	$B=0$, PGM $L=56$, none, Nesterov	0 (0)	54.99
Multi-channel dict (R1)	color	$B=K$, ADMM, SM, $C_d=C$	1.3×10^{-5}	279.66
Single-channel dict (R2)	color	$B=CK$, ADMM, SM, $C_d=1$	3.4×10^{-16}	330.24
Grouped consensus	gray	$B=2$, ADMM $\sigma=2.2$, ISM, $C_d=1$	collapse-checked (new)	82.81
collapse $B=1$	gray	as ISM	7.4×10^{-10} (3.4×10^{-8})	82.68
collapse $B=K$	gray	as consensus	2.5×10^{-10} (9.2×10^{-9})	83.11

Table 3: Equivalences between cells, re-derived from the axis structure (Sec. 5.2). These are identities between updates, not separate runs, so they carry verification evidence rather than a reference-matching deviation.

Equivalence	Cell identified with	Verification
“3D” = consensus	$B=K$, ADMM, SM	Block-circulant 3D objective equals the consensus objective $\sum_k \frac{1}{2} \ X_k \mathbf{g} - \mathbf{s}_k\ _2^2$ to relative error 0 for $K \in \{2, 3, 5\}$ (numeric); block DFT diagonalizes the operator.
Tiling = rank-1	rank-1 corner ($K'=1$)	Folding the K images into one tiled image degenerates the partition axis to a single rank-1 system; same cell as $B=K$ at $K=1$ (structural).
$B=K$ with ISM/CG inner	$B=K$, SM	Iterated SM with a single rank-1 term is literally one Sherman–Morrison step; a diagonal-plus-rank-1 system has two distinct eigenvalues, so CG returns the same solution (algebraic identity).

multi-channel-dictionary variants of the $B=1$ and grouped rows — that the template predicts are feasible but that neither the review nor the reference implementation instantiate.

8 Failure analysis and limitations

The abstraction is incomplete in three specific ways, which we report rather than smooth over. (i) *PGM is a limit, not a block*. FISTA sits at the degenerate $B=0$ corner of the partition axis: it shares the template’s projection and Nesterov skeleton, but it has no consensus blocks and its data-fidelity is handled by a gradient/MM step rather than a linear solve, so it does not participate in the rank- $|G|$ structure that organizes the ADMM cells. The partition axis is thus an ADMM-family axis; the linearization axis (A2) is what separates the two families. (ii) *CG is inexact*. The CG cell recovers the trajectory to 4.6×10^{-5} but the converged dictionary only to 5.2×10^{-3} at the reference’s 10^{-3} stopping tolerance; the error is a property of the inner solver’s tolerance, not of the template, and vanishes (2.6×10^{-8}) when the tolerance is tightened on both sides. (iii) *The inter-channel joint-sparsity representation is out of scope*. Coupling per-channel maps by a joint-sparsity penalty lives in the sparse-coding subproblem, which we hold fixed; the template therefore does not express it as a dictionary-update cell, and we do not claim it as a recovered representation. Finally, our recovery guarantees are established at small scale ($K=4$, $N=96$) and our multi-channel convergence runs are at a single color scale; the selection map’s quantitative thresholds are inferred from the linear-system structure and a $K=8$ sweep, not from an exhaustive (C , K) grid.

Table 4: The design space as a grid over the partition axis (A1, rows) and the inner-solver axis (A3, columns) at ADMM linearization (A2); the last row is the $B=0$ PGM limit, which has no linear solve. Cell marks: **R** = recovered and validated against SPORCO (Table 2); **N** = new cell instantiated here; **E** = equivalent to another cell (Table 3); *empty* = feasible but not instantiated; $\times$ = structurally undefined (rank mismatch). The channel axis (A4) overlays every cell: it is instantiated for the consensus row (R1 with $C_d=C$, R2 with $C_d=1$ folded) and remains *empty* for the other rows (e.g. grouped consensus with $C_d=C$).

A1 partition \ A3 inner solver	SM	ISM	CG
$B=1$ (rank K)	$\times$	ISM R	CG R
$1 < B < K$ (rank $ G $)	$\times$	grouped N	<i>empty</i>
$B=K$ (rank 1)	consensus R ; “3D” E	$\equiv$ SM E	$\equiv$ SM E
$B=0$ (PGM, no solve)	FISTA R (Nesterov, A5)		

9 Related work

The dictionary updates we abstract are due to Garcia-Cardona and Wohlberg [2018], building on efficient convolutional sparse coding Wohlberg [2014, 2016b], the subproblem coupling of Garcia-Cardona and Wohlberg [2017], the fast consensus update of Šorel and Šroubek [2016], and proximal-gradient convolutional coding Chalasani et al. [2013], Beck and Teboulle [2009]. The multi-channel dictionary model is from Wohlberg [2016a]; the review’s Sec. VI already carries the four updates to C channels and states the $K \leftrightarrow C$ duality, and the authors’ reference implementation Wohlberg [2020] ships color CDL examples. Our contribution is not the multi-channel extension itself — that is the review’s own Sec. VI, and reproducing it is baseline work here — but the *unification*: a single operator-splitting/MM derivation whose axes generate the separately-derived updates and the multi-channel representations, whose empty cells generate a new grouped-consensus update, and whose shared basis structure makes the “3D”/consensus equivalence a corollary rather than a coincidence. The MM perspective we use for the linearization axis is standard Hunter and Lange [2004]; convolutional (deconvolutional) models originate in Zeiler et al. [2010]. We are not aware of prior work that exposes the dictionary-update design space as a single template with a fillable intermediate-partition cell.

10 Conclusion

We have shown that the convolutional dictionary-update algorithms of Garcia-Cardona and Wohlberg are instances of one block-consensus operator-splitting/MM template, controlled by five near-orthogonal axes. One shared driver recovers the ISM, CG, consensus, and FISTA updates to within 10^{-3} (the exact solves to machine precision) against the reference implementation, and instantiates the two multi-channel representations to convergence on color data. Because the template is a genuine derivation, its design space predicts more than it was built from: a grouped/mesoscale consensus update that interpolates between the rank- K and rank-one methods and is validated by degenerate collapse, and the “3D”/consensus equivalence re-derived from the axis structure. The result is a design space rather than a list of algorithms, and a selection map that says which point in that space suits which (C, K) regime.

A Reproducibility

All quantitative claims trace to logs under proposal/logs/. The template and harnesses are in proposal/code/: `template.py` (the shared driver), `driver.py` (the dictionary-learning coupling, holding the sparse-coding stage fixed to SPORCO’s ADMM CBPDN), `run_recovery.py`, `run_multichannel.py`, `run_reveal.py`, and `run_characterize.py`; figures are regenerated from the logs by `make_figures.py`. The pre-registered framework (axes, cells, and the planned reveal) is in proposal/code/framework_plan.md, timestamped before the validation runs. References are computed by running SPORCO at runtime on identical inputs; no reference trajectory or dictionary is hardcoded. The design axes and recovered cells were fixed in the plan file before the runs and were not changed after seeing results.

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